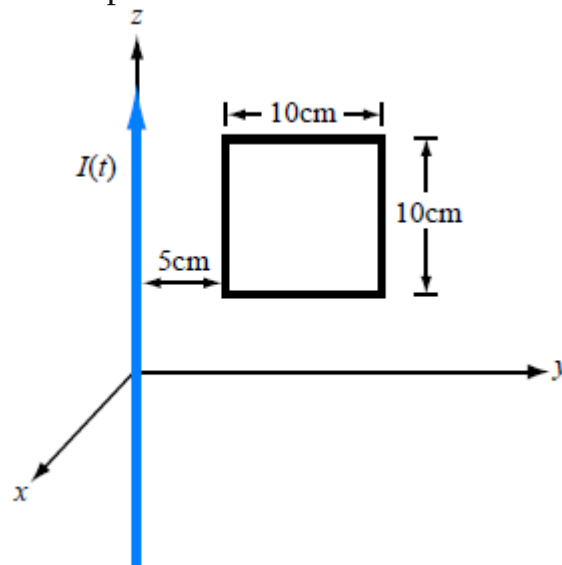


EE 340 Electromagnetic Theory

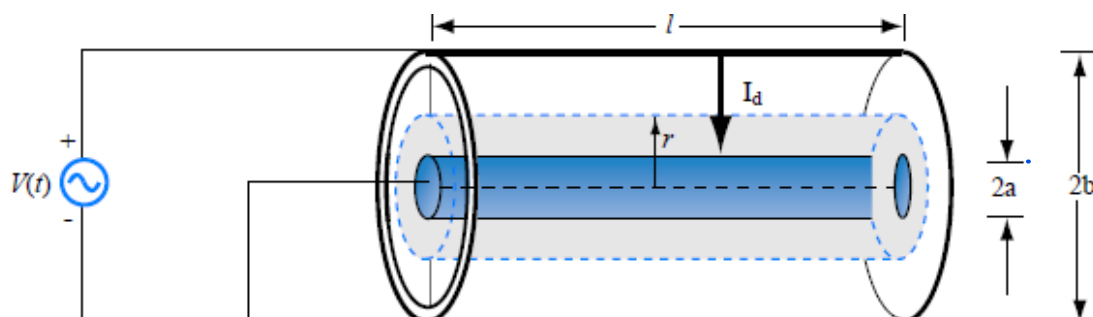
Assignment #02

Q.1 A long cylindrical conductor whose axis is coincident with the z -axis has a radius a and carries a current characterized by a current density $J = J_0 e^{-r} \hat{z}$, where J_0 is a constant and r is the radial distance from the cylinder's axis. Obtain an expression for the magnetic field H for (a) $0 \leq r \leq a$ and (b) $r > a$.

Q.2 The square loop shown in Figure is coplanar with a long, straight wire carrying a current $I(t) = 5 \cos(2\pi \times 10^4 t) \text{ A}$,
(a) Determine the emf induced in the loop.
(b) Determine the direction and magnitude of the current that would flow through the loop, if the loop has an internal resistance of 1Ω .



Q.3 A coaxial capacitor of length $l = 6 \text{ cm}$ uses an insulating dielectric material with $\epsilon_r = 9$. The radii of the cylindrical conductors are $a = 0.5 \text{ cm}$ and $b = 1 \text{ cm}$. If the voltage applied across the capacitor is $V(t) = 50 \sin(120\pi t)$, then what is the displacement current?



✓ Q.4 If the current density in a conducting medium is given by:

$$J(x, y, z, t) = (z^2 \hat{x} - 4y^2 \hat{y} + 2x \hat{z}) \cos(\omega t)$$

Determine the corresponding charge distribution ρ_v .

Q.5 Find out the generalized expression of frequency at which the conduction current density will be equal in magnitude to the displacement current density in a lossy dielectric medium?

✓ Q.6 Which of the following two materials is the better insulator?

Dry Soil: $\epsilon_r = 2.5$, $\sigma = 10^{-4} \text{ S/m}$

Fresh Water: $\epsilon_r = 80$, $\sigma = 10^{-3} \text{ S/m}$

✓ Q.7 The electric field of a plane wave propagating in a **nonmagnetic** material is given by:

$$E = 3 \sin(\pi \times 10^7 t - 0.2\pi x) \hat{y} + 4 \cos(\pi \times 10^7 t - 0.2\pi x) \hat{z} \text{ V/m}$$

Determine (a) the wavelength, (b) dielectric constant of the medium, (c) intrinsic impedance of the medium, and (d) the magnetic field H .

Also plot the 3 dimensional E field in matlab and create an animation of the E-field propagation.

Q.8 The magnetic field in a given dielectric medium ($\mu_r = 1$) is given by:

$$H = 6 \cos(2z) \sin(2 \times 10^7 t - 0.1x) \hat{y} \text{ A/m}$$

where x and z are in meters. Determine: (a) the electric field E , (b) the displacement current density, and (c) the charge density.

Also plot the 3 dimensional H field in matlab and create an animation of the H-field propagation.